

Research Paper on Distributed Systems

conditions. Future work will focus on geo-replicated deployments and Byzantine fault tolerance.

1 Introduction

Distributed systems are computer systems that consist of multiple independent components that communicate and coordinate their actions by passing messages. The components interact with each other in order to achieve a common goal. This paper explores the fundamental concepts and challenges in designing reliable distributed systems.

2 Background

2.1 Consensus Algorithms

Consensus algorithms are fundamental to distributed systems. They ensure that multiple nodes agree on a single value. The Paxos algorithm is one of the most well-known consensus protocols. Raft is a more recent alternative that is easier to understand and implement.

2.2 Data Flow

Data flow in distributed systems follows specific patterns. The most common patterns include publish-subscribe and request-response. In publish-subscribe, producers send messages to topics without knowing the specific consumers. In request-response, a client sends a request and waits for a response from the server.

3 Methodology

We conducted experiments on a cluster of 100 nodes. Each node was equipped with 64GB RAM and 16 CPU cores. The network had a round-trip latency of 1 millisecond and a bandwidth of 10 gigabits per second. We measured throughput, latency, and fault tolerance under various failure scenarios.

4 Results

The results show that our proposed algorithm achieves 99.9 percent availability under normal conditions. The latency remains below 50 milliseconds for 95 percent of requests. Under failure conditions with up to 30 percent of nodes failing, the system continues to operate with slightly degraded performance. The consensus protocol converges within 200 milliseconds in all tested scenarios.

5 Conclusion

We have demonstrated that our distributed consensus algorithm provides high availability and low latency under both normal and failure